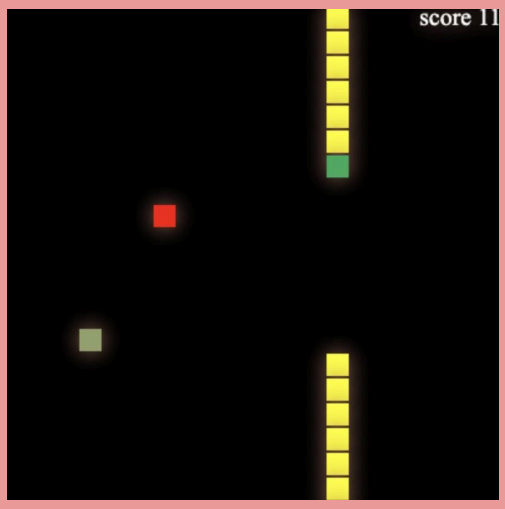
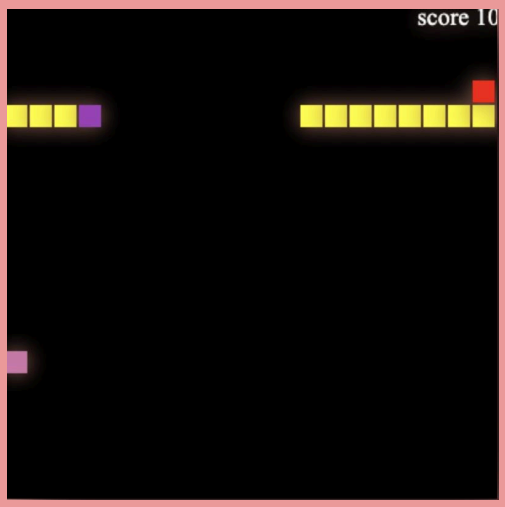
Project report

Development Description :

The project is about a traditional snake game that has been enhanced with new features. The game was based on a YouTube tutorial, but I've added several new features to make it more engaging such as :

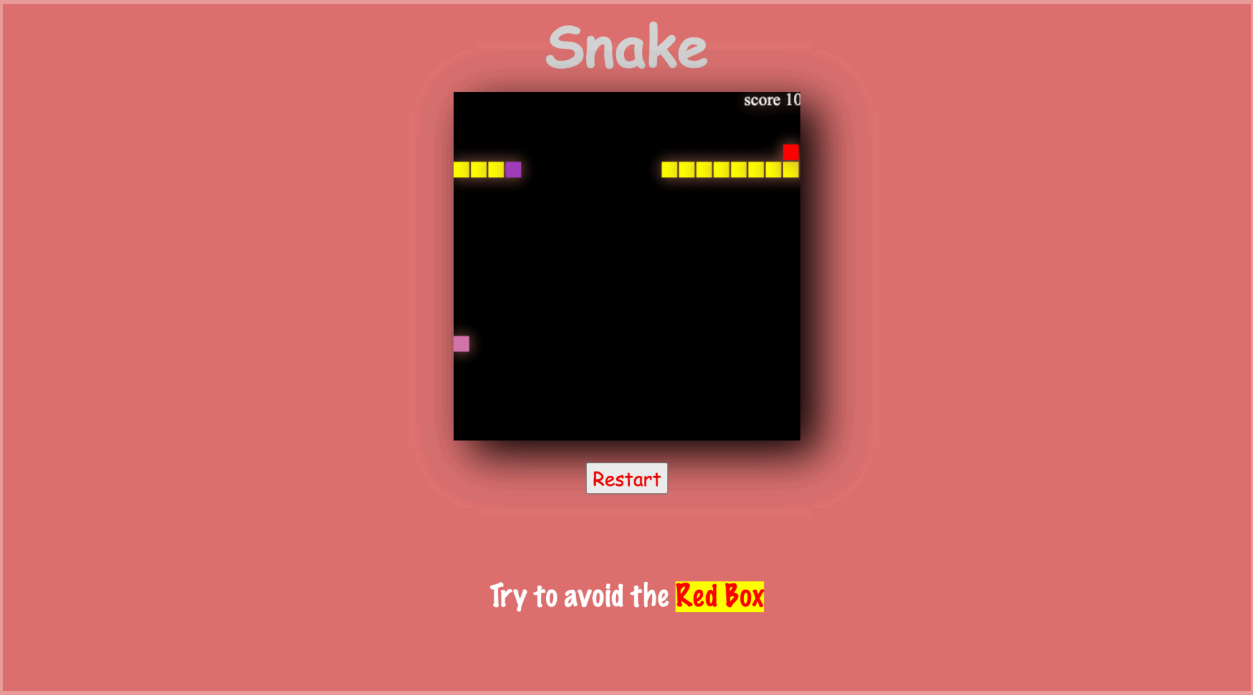
- The first change made to the game was the boundary. Instead of having a fixed boundary, the game has been designed to have a looped boundary. This means that if the snake hits the left wall, it will move to the right wall, and vice versa.





- I have designed the game with different colors and styles to make it more visually appealing. These changes enhance the game's overall presentation and make it more enjoyable to play.

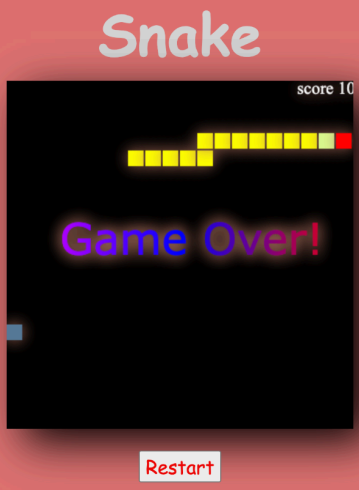
- I made the game more challenging by introducing a bomb or a red box that appears in different places inside the canvas whenever the snake eats an apple. The player needs to avoid the bomb, which adds an extra level of difficulty to the game.



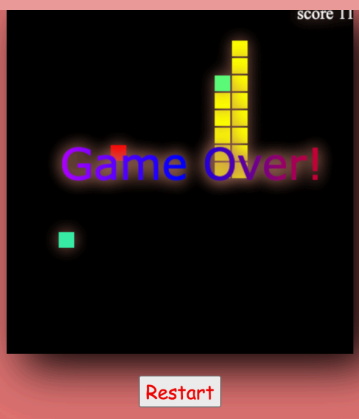
- To make it easier for players to start again after losing, I've added a Restart button . This button refreshes the page and restarts the game, allowing players to quickly try again.

- As the player progresses through the game, the speed of the snake increases when the player reaches certain milestones. The snake's speed increases when the player scores 10, 20, and 30 apples.

- I have also added a feedback feature to the game. When the game is over, under the canvas, the player is given information on their score and why the game ended. This information helps the player understand what they did wrong and how they can improve their gameplay.



the snake collided with the Red box, You gained 10 apples! Click Restart to play again



the snake collided with its own body, You gained 11 apples! Click Restart to play again

User Manual

- Use the arrow keys to move the snake around the screen.
- Eat the apples that appear on the screen with auto generated colors that is simmilar to the snake's head to increase your score.
- Avoid the red bomb or box that appears on the screen when the snake eats an apple.
- If the snake touches the boundary of the screen, it will loop around to the other side.
- If the snake touches its own body, the game will end.
- When the game ends, you can click the "Restart" button to start over.

It's important to remember that the snake's body will grow as it eats apples, making it harder to navigate around the screen. The red bomb or box adds an extra level of difficulty, and the looping boundary can be tricky to get used to. Be sure to keep an eye on the score and watch out for the bomb!

References which I used to built this Game :

Youtube tutorial:

- https://www.youtube.com/watch?v=baBq5GAL0_U&t=1211s

Websites :

- https://www.w3schools.com/
- https://developer.mozilla.org/en-US/
- https://tailwindcss.com/docs/background-color

Most of the functions and the code was written based on the examples from the slides and the previous work from the labs.